3.12 Δ-Graded Rings

References:

- [Bou98a] (Chap. II §11)

Let Δ be a commutative monoid. By convention we write the monoid operation additively and 0 for the identity. Further we assume it is **cancellable**: $\lambda + \mu = \lambda + \mu' \implies \mu = \mu'$. Typically $\Delta = \mathbb{Z}$, but for example we may also consider $\mathbb{N}^k$ for some positive integer k.

Definition 3.12.1 (Graded Abelian Groups)
*Let Δ be a cancellable commutative monoid. An abelian group G is Δ-**graded** if it may be written as internal direct sum (3.3.92) of $\mathbb{Z}$-modules*

$$G = \bigoplus_{\lambda \in \Delta} G_\lambda$$

for subgroups $G_\lambda \leq G$. In particular the subgroups G_λ are disjoint. This means for every $g \in G$ there is a unique decomposition into a finite sum

$$g = \sum_{\lambda \in \Delta} g_\lambda$$

*If $g \in G_\lambda$ then we say g is **homogenous**. In this case we write $\delta(g) \coloneqq \lambda$ for the **degree** of g.*

We may write $\pi_\lambda : G \to G_\lambda$ and $i_\lambda : G_\lambda \to G$ for projection and inclusion respectively. For $g \in G$ we may abbreviate $\pi_\lambda(g)$ by g_λ.

Definition 3.12.2 (Graded Ring)
*Let Δ be a cancellable commutative monoid. A commutative ring A is **graded** of type Δ if the additive group $(A, +)$ is Δ-graded, and the multiplication is compatible in the sense that*

$$a \in A_\lambda, b \in A_\mu \implies a \cdot b \in A_{\lambda+\mu}$$

Lemma 3.12.3
Let A be a Δ-ring. Then A_0 is a subring.

In the case $\Delta = \mathbb{N}$ the additive subgroups

$$A_{\geq k} \coloneqq \sum_{l \geq k} A_l$$

*are homogenous ideals. We write $A_+ \coloneqq A_{\geq 1}$ which we call the **irrelevant ideal**.*

Proof. We need only show $1 \in A_0$. By hypothesis there is a unique decomposition

$$1 = \sum_{\lambda \in \Delta} e_\lambda \quad e_\lambda \in A_\lambda$$

For every $x \in A_\mu$ we have

$$x = x \cdot 1 = \sum_{\lambda \in \Delta} x \cdot e_\lambda$$

By the cancellable assumption we have $\lambda \neq 0 \implies \lambda + \mu \neq \mu$. Therefore $x \cdot e_0 = x$. By distributivity this then holds for all $x \in A$ and in particular $1 = 1 \cdot e_0 = e_0$. ∎

Definition 3.12.4 (Graded Module)
*Let A be a Δ-graded ring and M a left (resp. right) A-module which is a Δ-graded as an abelian group. Then M is a **graded module** if the A-module structure is compatible with the grading, namely*

$$a \in A_\lambda, m \in M_\mu \implies a \cdot m \in M_{\lambda+\mu}$$

Elements of M_λ are known as homogenous.

Definition 3.12.5 (Graded Homomorphisms)
*Let A, B be Δ-graded rings and $\phi : A \to B$ a ring homomorphism. It is a **graded ring homomorphism** if*

$$\phi(A_\lambda) \subseteq B_\lambda \quad \forall \lambda \in \Delta$$

Let M, M' be graded A-modules and $\phi : M \to M'$ an A-module homomorphism. Then ϕ is a ***graded module homomorphism*** *if*

$$\phi(M_\lambda) \subseteq M'_\lambda \quad \forall \lambda \in \Delta$$

Definition 3.12.6 (Graded Submodules)
Let M be a graded A-module, and $N \leq M$ an A-submodule. Then the following are equivalent

a) $n \in N, \lambda \in \Delta \Longrightarrow \pi_\lambda(n) \in N$
b) N is generated by homogenous elements
c) N is equal to the internal direct sum

$$\bigoplus_{\lambda \in \Delta} N \cap M_\lambda$$

We say such a submodule is ***graded*** *or* ***homogenous*** *and we write $N_\lambda \coloneqq N \cap M_\lambda$.*

Proposition 3.12.7
Let M be a graded A-module and $\{N_\alpha\}_{\alpha \in I}$ be a family of submodules. Then both

$$\sum_{\alpha \in I} N_\alpha$$

and

$$\bigcap_{\alpha \in I} N_\alpha$$

are graded submodules.

Proposition 3.12.8
Let M be a finitely-generated graded A-module. Then M has a finite generating set of homogenous elements.

Definition 3.12.9 (Homogenous Ideal)
Let A be a commutative graded ring. An ideal $\mathfrak{a} \lhd A$ which is homogenous as an A-submodule of A is a ***homogenous ideal****. We write $\mathfrak{a}_\lambda \coloneqq \mathfrak{a} \cap A$, then by definition*

$$\mathfrak{a} = \bigoplus_{\lambda \in \Delta} \mathfrak{a}_\lambda$$

Note if A is Noetherian then every homogenous ideal has a finite generating set of homogenous elements.

Lemma 3.12.10
Let $\phi : A \to B$ be a graded ring and $\mathfrak{b} \lhd B$ a homogenous ideal. Then $\phi^{-1}(\mathfrak{b})$ is homogenous.

Example 3.12.11 (Polynomial Ring)
The polynomial ring $k[X_0, ..., X_n]$ is a graded ring over $\mathbb{N}$ with

$$k[X_0, ..., X_n]_d = k\langle \{ X_1^{\alpha_1} \cdot ... \cdot X_n^{\alpha_n} \mid \alpha_1 + ... + \alpha_n = d \} \rangle$$

Furthermore

$$\bigoplus_{i \geq d} k[X_0, ..., X_n]_i = (X_0, ..., X_n)^d$$

Proposition 3.12.12 (Quotient by a Homogenous Ideal)
Suppose Δ is a commutative cancellable monoid. Let A be a Δ-graded ring and $\mathfrak{a} \lhd A$ a homogenous ideal. Then $A/\mathfrak{a}$ is a Δ-graded ring by considering the internal direct sum of abelian groups

$$A/\mathfrak{a} = \bigoplus_{\lambda \in \Delta} (A/\mathfrak{a})_\lambda$$
$$(A/\mathfrak{a})_\lambda \coloneqq \{ \overline{a} \mid a \in A_\lambda \}$$

Furthermore there is an isomorphism of A-modules

$$(A/\mathfrak{a})_\lambda \xrightarrow{\sim} A_\lambda/(\mathfrak{a} \cap A_\lambda)$$

For every homogenous ideal $\mathfrak{b} \supseteq \mathfrak{a}$ the quotient ideal $\mathfrak{b}/\mathfrak{a}$ is homogenous with

$$\mathfrak{b}/\mathfrak{a} = \bigoplus_{\lambda\in\Delta} (\mathfrak{b}/\mathfrak{a})_\lambda$$

Proof. Evidently the $(A/\mathfrak{a})_\lambda$ are abelian subgroups and span $A/\mathfrak{a}$. Observe that

$$0 = \sum_{\lambda\in\Delta} \overline{a_\lambda} \implies \sum_{\lambda\in\Delta} a_\lambda \in \mathfrak{a}$$

Then as $\mathfrak{a}$ is homogenous we conclude $a_\lambda \in \mathfrak{a}$ and therefore $\overline{a_\lambda} = 0$. Therefore the family $(A/\mathfrak{a})_\lambda$ satisfy the criteria of (3.3.93) and the direct sum is well-defined. Evidently the ring multiplication on $A/\mathfrak{a}$ is compatible with the given Δ-grading.

The isomorphism is induced by π_λ.

Suppose that $\overline{b} \in \mathfrak{b}/\mathfrak{a}$ then we require to prove that $\left(\overline{b}\right)_\lambda \in (\mathfrak{b}/\mathfrak{a})$. By assumption $b_\lambda \in \mathfrak{b}$ so $\overline{b_\lambda} \in \mathfrak{b}/\mathfrak{a}$. However by the first part $\overline{b_\lambda} = \left(\overline{b}\right)_\lambda$. ∎

Proposition 3.12.13 (Localization at Homogenous Elements)
Let Δ be an abelian group, A a Δ-graded ring and $S \subset A$ a multiplicative set of ***homogenous*** *elements. Then we may define a Δ-gradation on $S^{-1}A$ by*

$$\left(S^{-1}A\right)_\lambda := \left\{ \frac{a}{s} \;\middle|\; \delta(a) - \delta(s) = \lambda \right\}$$

For a homogenous ideal $\mathfrak{a} \lhd A$ then the ideal $S^{-1}\mathfrak{a}$ is homogenous.

Proof. We first show that $\left(S^{-1}A\right)_\lambda$ is an additive subgroup, for given homogenous $a, a' \in A$ and $s, s' \in s$ by definition

$$\frac{a}{s} + \frac{a'}{s'} = \frac{as' + a's}{ss'}$$

Suppose $\delta(a) - \delta(s) = \lambda = \delta(a') - \delta(s')$ then $\delta(as') = \delta(a's)$. Therefore $as' + a's$ is homogenous and $\delta(as' + a's) = \delta(as') = \lambda + \delta(s) + \delta(s') = \lambda + \delta(ss')$ whence $\left(S^{-1}A\right)_\lambda$ is an additive subgroup. Similarly for the multiplicative structure suppose $\delta(a) = \delta(s) + \lambda$ and $\delta(a') = \delta(s') + \lambda'$ then

$$\delta(aa') = \delta(a) + \delta(a') = \delta(\lambda) + \delta(\lambda') + (\lambda + \lambda') = \delta(\lambda\lambda') + (\lambda + \lambda')$$

so that $\frac{a}{s} \cdot \frac{a'}{s'} \in \left(S^{-1}A\right)_{\lambda+\lambda'}$ as required. Evidently $S^{-1}A$ is the internal sum of $\left(S^{-1}A\right)_\lambda$. To show it is direct suppose that

$$0 = \sum_{i=1}^{n} \frac{a_i}{s_i} \quad \delta(a_i) = \delta(s_i) + \lambda_i$$

for distinct $\lambda_1, ..., \lambda_n$. Then

$$0 = \sum_{i=1}^{n} a_i t \prod_{j\neq i} s_j$$

for some $t \in S$. Furthermore

$$\delta\left(a_i t \prod_{j\neq i} s_j\right) = \delta(a_i) + \sum_{j\neq i} \delta\left(s_j\right) + \delta(t) = \lambda_i + \sum_{j=1}^{n} \delta\left(s_j\right) + \delta(t)$$

As Δ is cancellable the elements $\hat{\lambda}_i := \lambda_i + \sum_{j=1}^{n} \delta\left(s_j\right) + \delta(t)$ are distinct. By assumption we conclude that $a_i t \prod_{j\neq i} s_j = 0$ and so $\frac{a_i}{s_i} = 0$. Therefore $S^{-1}A$ is the direct sum of $\left(S^{-1}A\right)_\lambda$ by (3.3.93).

For $\frac{a}{s} \in S^{-1}\mathfrak{a}$ we wish to show that $\left(\frac{a}{s}\right)_\lambda \in S^{-1}\mathfrak{a}$ for all $\lambda \in \Delta$. Here $a \in \mathfrak{a}$ so by assumption $a_\lambda \in \mathfrak{a}$ for all $\lambda \in \Delta$ and $a = \sum_{\lambda\in\Delta} a_\lambda$ so

$$\frac{a}{s} = \sum_{\lambda\in\Delta} \frac{a_\lambda}{s} = \sum_{\lambda\in\Delta} \frac{a_{\lambda+\delta(s)}}{s}$$

and so $\left(\frac{a}{s}\right)_\lambda = \frac{a_{\lambda+\delta(s)}}{s} \in S^{-1}\mathfrak{a}$ as required. ∎

Recall for any Δ-graded ring A the additive subgroup A_0 is also a subring.

Definition 3.12.14
Let Δ be a commutative abelian group, A a Δ-graded ring and S be a multiplicative subset of homogenous elements. Define the subring of $S^{-1}A$

$$A_{(S)} \coloneqq \left(S^{-1}A\right)_0 = \left\{ \frac{a}{s} \;\middle|\; \delta(s) = \delta(a) \right\}$$

Similarly for $\mathfrak{a}$ a homogenous ideal define the homogenous ideal of $S^{-1}A$

$$\mathfrak{a}_{(S)} \coloneqq \left\{ \frac{a}{s} \;\middle|\; a \in \mathfrak{a}, s \in S, \delta(s) = \delta(a) \right\}$$

In the case $\mathfrak{p}$ is a homogenous prime ideal define the local ring

$$A_{(\mathfrak{p})} \coloneqq A_{(\{x \notin \mathfrak{p}\})}$$

and for $f \in A$ homogenous define

$$A_{(f)} \coloneqq A_{(\{1, f, f^2, \ldots\})}$$

Lemma 3.12.15
Let Δ be an abelian group, A a Δ-graded ring and $\mathfrak{a} \lhd A$ a homogenous ideal. Then

$$\mathfrak{a}_{(S)} = A_{(S)} \cap S^{-1}\mathfrak{a}$$

as subsets of $S^{-1}A$.

Lemma 3.12.16
Let Δ be an abelian group, A a Δ-graded ring, $a \in A_\mu$ homogenous and $x \in A$. Then

$$(ax)_\lambda = a \cdot x_{\lambda - \mu}$$

In particular ax is homogenous iff x is homogenous.

Proposition 3.12.17 (Homogenous localisation comutes with quotients)
Let Δ be an abelian group, A a Δ-graded ring, $\mathfrak{a}$ a homogenous ideal and $S \subset A$ a multiplicatively closed set of homogenous elements such that $S \cap \mathfrak{a} = \varnothing$. Then there is a canonical isomorphism of Δ-graded rings

$$\begin{aligned} (S^{-1}A)/(S^{-1}\mathfrak{a}) &\xrightarrow{\sim} \pi(S)^{-1}(A/\mathfrak{a}) \\ \frac{a}{s} + S^{-1}A &\to \frac{a + \mathfrak{a}}{s + \mathfrak{a}} \end{aligned}$$

where $\pi : A \to A/\mathfrak{a}$ is the canonical surjection. In particular there is an isomorphism of rings

$$A_{(S)}/\mathfrak{a}_{(S)} \xrightarrow{\sim} \left(S^{-1}A/S^{-1}\mathfrak{a}\right)_0 \xrightarrow{\sim} (A/\mathfrak{a})_{(\pi(S))}$$

Proof. The first isomorphism of rings is from (3.6.25), and it is obviously compatible with the Δ-grading. For suppose $x \in (S^{-1}A)/(S^{-1}\mathfrak{a})$ with $\delta(x) = \lambda$. Then $x = \frac{a}{s}$ with $\delta(a) - \delta(s) = \lambda$. By definition $\delta(a + \mathfrak{a}) - \delta(s + \mathfrak{a}) = \lambda$ as required.

The third isomorphism follows immediately. For the second consider the ring homomorphism

$$\phi : A_{(S)} \hookrightarrow S^{-1}A \to S^{-1}A/S^{-1}\mathfrak{a}$$

Then evidently the image is $\left(S^{-1}A/S^{-1}\mathfrak{a}\right)_0$ as the quotient map is graded. The kernel is $A_{(S)} \cap S^{-1}\mathfrak{a}$ which equals $\mathfrak{a}_{(S)}$ by (3.12.15). ∎

Proposition 3.12.18
Let Δ be an abelian group, A a Δ-graded ring, $\mathfrak{a}$ a homogenous ideal and $\mathfrak{p} \supseteq \mathfrak{a}$ a homogenous prime ideal. Then there is a canonical isomorphism of Δ-graded rings

$$\begin{aligned} A_{\mathfrak{p}}/\mathfrak{a}_{\mathfrak{p}} &\xrightarrow{\sim} (A/\mathfrak{a})_{\mathfrak{p}/\mathfrak{a}} \\ \frac{a}{s} + \mathfrak{a}_{\mathfrak{p}} &\to \frac{a + \mathfrak{a}}{s + \mathfrak{a}} \end{aligned}$$

In particular there is an isomorphism of rings

$$A_{(\mathfrak{p})}/\mathfrak{a}_{(\mathfrak{p})} \xrightarrow{\sim} \left(A_{\mathfrak{p}}/\mathfrak{a}_{\mathfrak{p}}\right)_0 \xrightarrow{\sim} (A/\mathfrak{a})_{(\mathfrak{p}/\mathfrak{a})}$$

given by

$$\frac{a}{s} + \mathfrak{a}_{(\mathfrak{p})} \to \frac{a+\mathfrak{a}}{s+\mathfrak{a}}$$

Proof. Follows from (3.12.17) and the observation that $A/\mathfrak{a} \setminus \mathfrak{p}/\mathfrak{a} = \pi(A \setminus \mathfrak{p})$. ∎